

Table 3 Velocity as a function of time.

t (s)	$v(t)$ (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

- Determine the value of the velocity at $t = 16$ seconds using quadratic splines.
- Using the quadratic splines as velocity functions, find the distance covered by the rocket from $t = 11$ s to $t = 16$ s.
- Using the quadratic splines as velocity functions, find the acceleration of the rocket at $t = 16$ s.

Solution

a) Since there are six data points, five quadratic splines pass through them.

$$\begin{aligned}
 v(t) &= a_1 t^2 + b_1 t + c_1, \quad 0 \leq t \leq 10 \\
 &= a_2 t^2 + b_2 t + c_2, \quad 10 \leq t \leq 15 \\
 &= a_3 t^2 + b_3 t + c_3, \quad 15 \leq t \leq 20 \\
 &= a_4 t^2 + b_4 t + c_4, \quad 20 \leq t \leq 22.5 \\
 &= a_5 t^2 + b_5 t + c_5, \quad 22.5 \leq t \leq 30
 \end{aligned}$$

The equations are found as follows.

1. Each quadratic spline passes through two consecutive data points.

$a_1 t^2 + b_1 t + c_1$ passes through $t = 0$ and $t = 10$.

$$a_1(0)^2 + b_1(0) + c_1 = 0 \quad (1)$$

$$a_1(10)^2 + b_1(10) + c_1 = 227.04 \quad (2)$$

$a_2 t^2 + b_2 t + c_2$ passes through $t = 10$ and $t = 15$.

$$a_2(10)^2 + b_2(10) + c_2 = 227.04 \quad (3)$$

$$a_2(15)^2 + b_2(15) + c_2 = 362.78 \quad (4)$$

$a_3 t^2 + b_3 t + c_3$ passes through $t = 15$ and $t = 20$.

$$a_3(15)^2 + b_3(15) + c_3 = 362.78 \quad (5)$$

$$a_3(20)^2 + b_3(20) + c_3 = 517.35 \quad (6)$$

$a_4 t^2 + b_4 t + c_4$ passes through $t = 20$ and $t = 22.5$.

$$a_4(20)^2 + b_4(20) + c_4 = 517.35 \quad (7)$$

$$a_4(22.5)^2 + b_4(22.5) + c_4 = 602.97 \quad (8)$$